
INTENT-BASED "VIBE"
ITERATION,
SPEC-DRIVEN AND
AGENTIC ORCHESTRATION

LEARNING GOALS

After this lesson, you will be able to:



INTENT BASED

Telling the system what you want



SPEC BASED

Telling the system, the exact specifications of the sytem

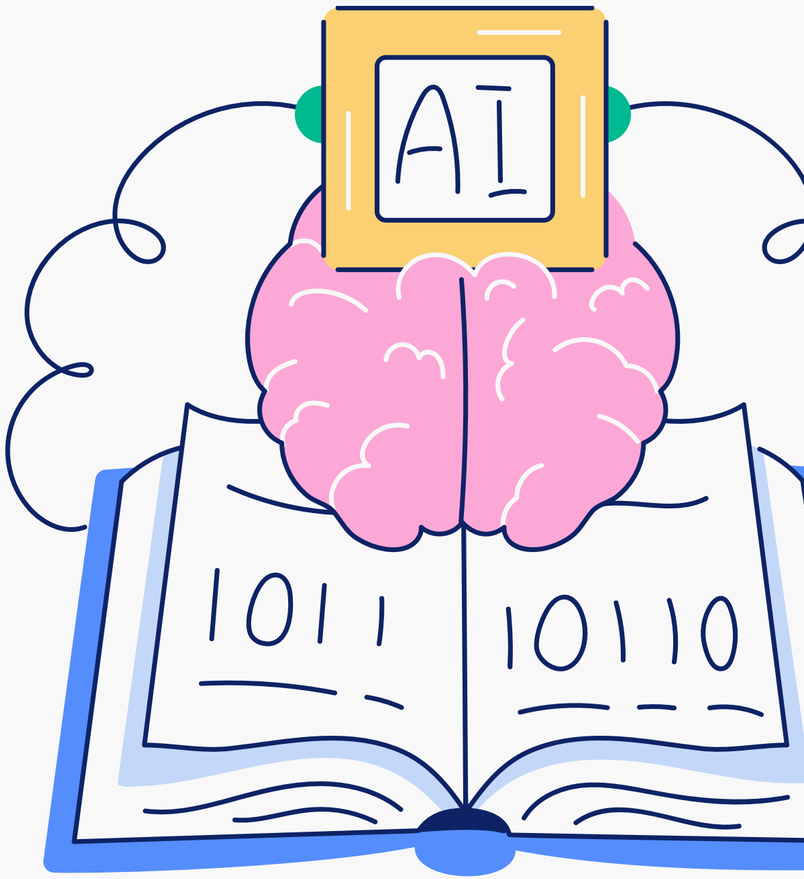


AGENTIC ORCHESTRATION

Managing agents, and ensuring there is no “clash of ideas”

NORMAL PROMPT

✗ BUILD A DASHBOARD

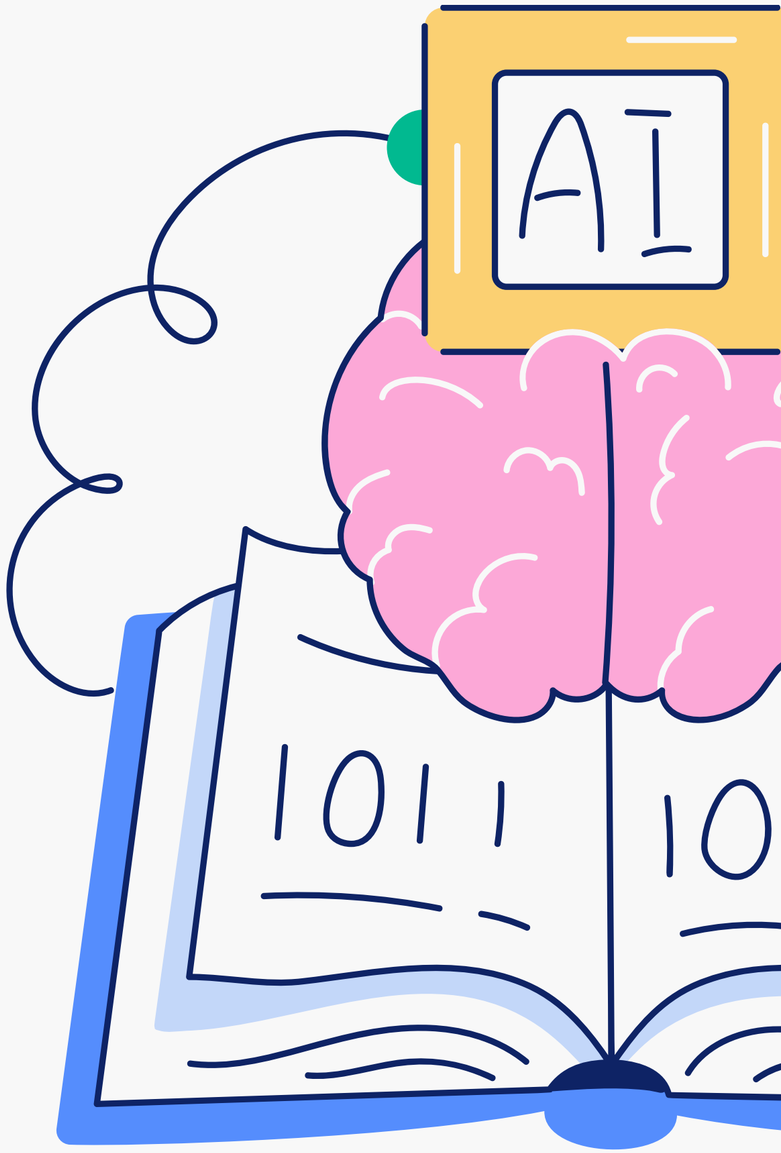


- Dashboard is built - Not the right colour
- Dashboard is built - In a language you don't understand
- Dashboard is built - Data shown hallucinates
- Dashboard is built - UI doesn't make sense

AND THEN YOU.. BURN TOKENS

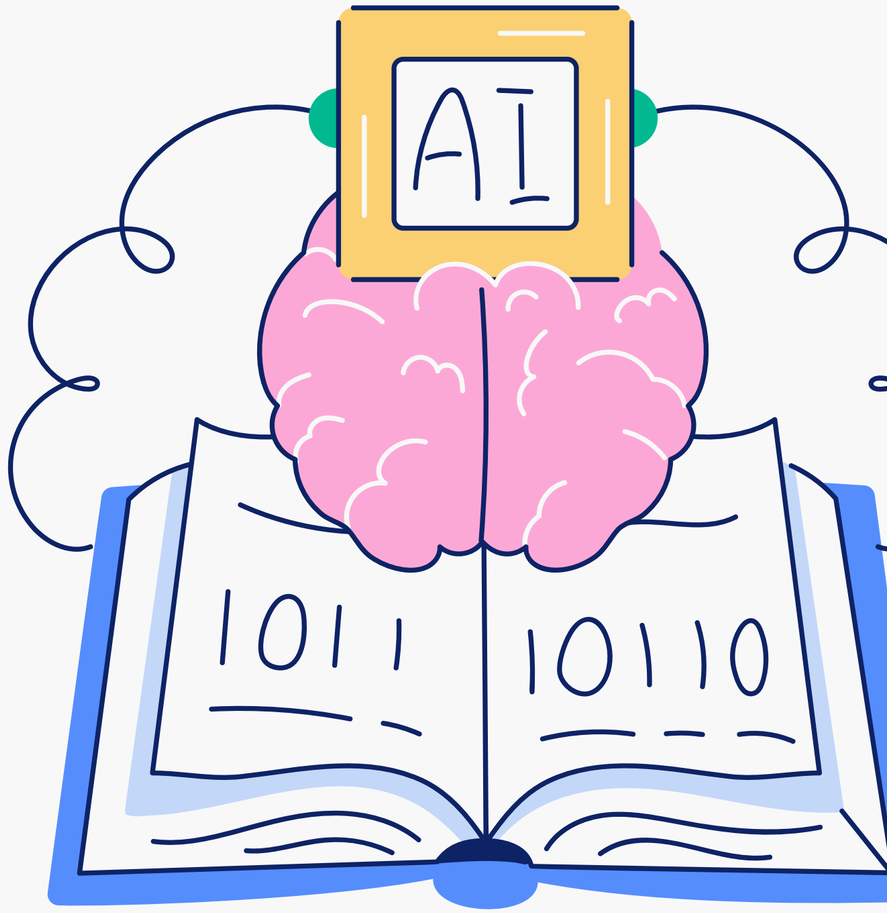
VIBE CODING

- ❌ RANDOM CODING
- ❌ BURNING TOKENS
- ❌ WASTING EARTH'S WATER



"INTENT" BASED DEV

- Start with intent - Describing exactly what you want
- Starts with derived outcome, rather than implementation details
- Describe the desired outcome rather than implementation details.



BE CLEAR IN WHAT THE APP DOES

"BUILD ME A AUTHENTICATION SYSTEM THAT WORKS"



AI can generate something.

- But what does works mean?
- Authentication requirements?
- Who is allowed?
- Idempotency?
- Authentication?
- Failure recovery?
- Latency?
- Data retention?
- Security...?

This leads naturally to spec-driven development.



SPEC-DRIVEN

SPEC-DRIVEN DEVELOPMENT MEANS DEFINING AN EXPLICIT, MACHINE-READABLE OR HUMAN-READABLE SPECIFICATION OF WHAT THE SYSTEM SHOULD DO BEFORE OR ALONGSIDE IMPLEMENTATION.

THE SPECIFICATION BECOMES THE CONTRACT BETWEEN INTENT AND IMPLEMENTATION.

IT'S SPECIFIC, BASED ON OUR 5-STAR FORMULA

WHAT IS A GOOD SPEC?



FUNCTIONAL REQUIREMENTS

- What must the system do?



CONSTRAINTS

- What must it NOT do?



ERFACES

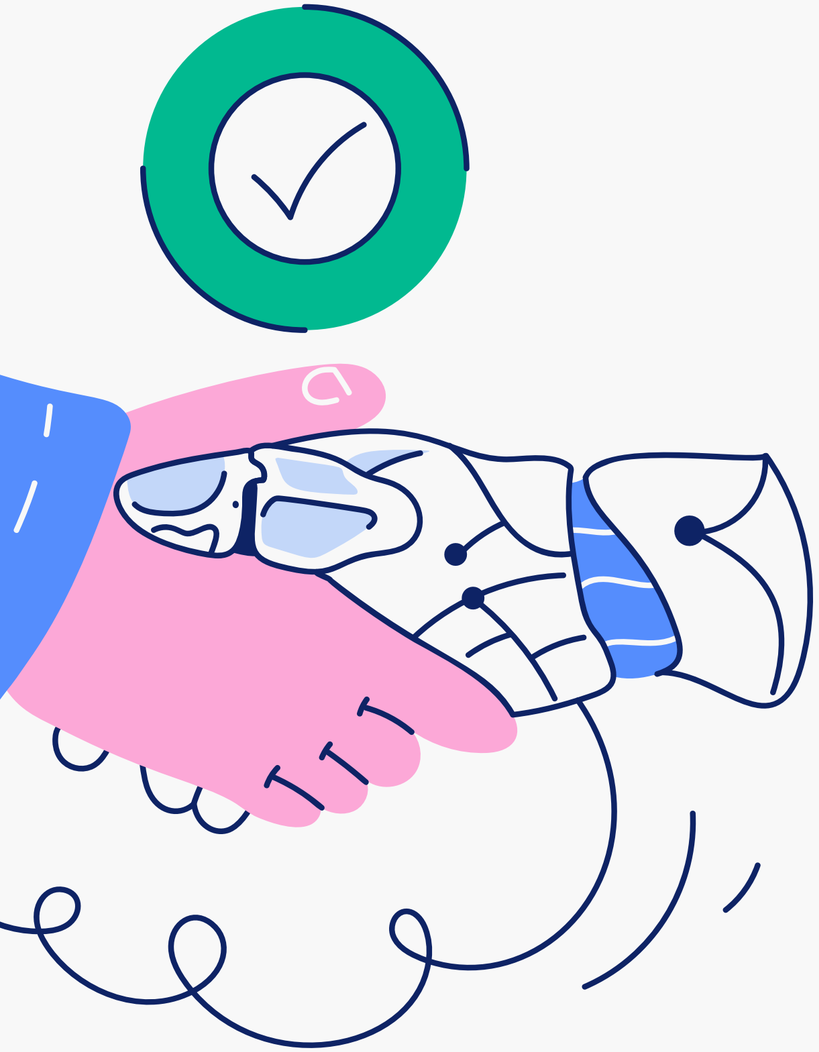
- How does it interact with other systems?



ACCEPTANCE CRITERIA

- How do we know it works?

WHAT IS A GOOD SPEC?



NON-FUNCTIONAL REQUIREMENTS

- security
- latency
- scalability
- availability
- cost
- compliance



BUILD A LOGIN SYSTEM

EG:

USER ENTERS EMAIL + PASSWORD

IF CREDENTIALS ARE VALID:
  CREATE AUTHENTICATED SESSION

IF CREDENTIALS ARE INVALID:
  RETURN GENERIC ERROR

AFTER 5 FAILED ATTEMPTS:
  TEMPORARILY LOCK AUTHENTICATION



AUTH SPECIFICATION

PASSWORD:
  MINIMUM 12 CHARACTERS

SESSION:
  EXPIRES AFTER 30 MINUTES OF INACTIVITY

ALL AUTHENTICATION EVENTS:
  MUST BE LOGGED



PROMPT → CODE

INTENT →

SPECIFICATION →



ACCEPTANCE CRITERIA →

TESTS → IMPLEMENTATION →

VERIFICATION

REQ: MY PROJECT'S REQUIREMENTS ARE A,B,C

ACCEPTANCE CRITERIA:

P, Q, R MEANS IT'S A SUCCESS



AGENTIC ORCHESTRATION

For tasks that get complex, all leading LLM's Claude, Gemini, GPT execute multiple agents that go and do the work. As vibe coders, we have to "manage" how these agents work



BREAKS INTO AGENTS

A MAIN AGENT BREAKS A TASK INTO PIECES AND CAN SPIN UP SUBAGENTS TO EXPLORE THE CODEBASE, RUN TESTS, OR RESEARCH SOMETHING. EACH SUBAGENT STARTS WITH A CLEAN CONTEXT WINDOW AND REPORTS BACK A SUMMARY,



CONTEXT WINDOWS

THE AGENT ONLY "KNOWS" WHAT'S IN ITS CONTEXT WINDOW: YOUR INSTRUCTIONS, FILES IT HAS READ, TOOL OUTPUTS, AND ITS OWN PRIOR REASONING. LONG SESSIONS DEGRADE AS THAT FILLS WITH STALE DETAIL.

AGENTIC ORCHESTRATION

For tasks that get complex, all leading LLM's Claude, Gemini, GPT execute multiple agents that go and do the work. As vibe coders, we have to "manage" how these agents work



FEEDBACK

WITHOUT A FEEDBACK SIGNAL, THEY TEND TO DECLARE SUCCESS TOO EARLY. WHAT YOU CAN DO: HAVE TESTS (EVEN ROUGH ONES) BEFORE ASKING FOR BIG CHANGES, TELL THE AGENT EXPLICITLY TO RUN THEM AND NOT STOP UNTIL THEY PASS, AND COMMIT OFTEN SO YOU CAN ROLL BACK WHEN IT GOES OFF THE RAILS.

PROJECT - BUILD A WEBSITE

BUILD Your own Portfolio, Design & Edit the Website. Later host it.

- Download VS Code, and Claude Code as an extention.
- Host website locally, make small changes to see how edits are done
- Learn about planning and execution mode in claude code
- Learn the different elements of a website designining






ACTIVITY LAB

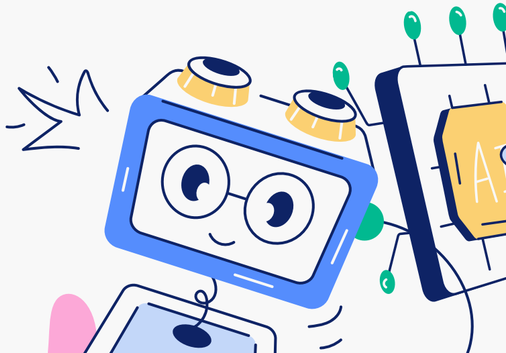
BRAINSTORM FEATURES: STATIC, SCROLL-WORTHY

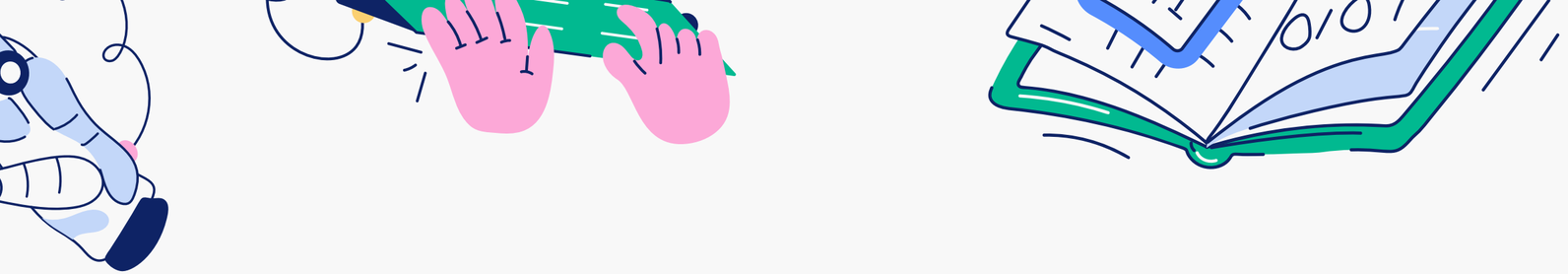
BRAINSTORM COLOURS: RED, BLUE

BRAINSTORM CONTENT: PREPARE CONTENT ON A PDF SHEET WITH CLEAR HEADERS

BRAINSTORM ANIMATION: WHAT MOVES, WHAT'S STATIC.








DID YOU THINK ABOUT?

FRONTEND?

BACKEND?

SCALING?

LINKING WEBSITE TO CLOUD?

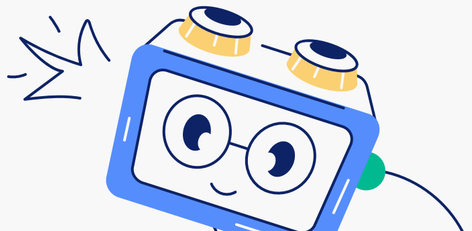
OPERARTIONS? DEVOPS?

SECURITY?

ATOMICITY?






DEMO
WEBSITE IN ACTION


AI
AI
AI

PROJECT - JOB SEEKER PORTAL

Use Cursor/Replot

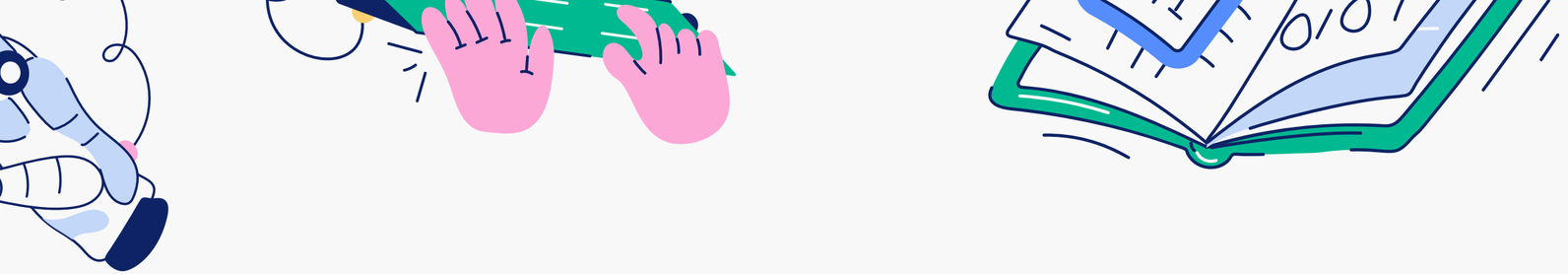
Don’t Prompt.

Use 5 Star formula.

Host website locally, make small changes to see how edits are done

Add features, links to 3-4 portal. Subscribe for auto-updates to user. Use MCP to link to gmail, or other docs

Include:
Scalability, Atomicity, Security , Code Quality, Functionality, Architecture, Performance






DID YOU THINK ABOUT?

FRONTEND?

BACKEND?

SCALING?

SEO?

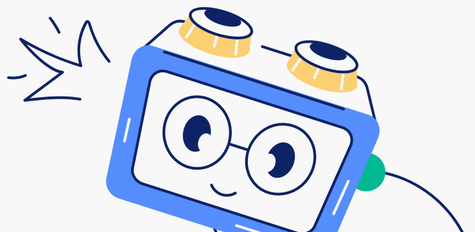
MONETISATION?

DATA MODEL?

MATCHING AND AI?






DEMO
JOB PORTAL IN ACTION


CHECK OUT A MVP:
HTTPS://WWW.VISTAARAAI.COM/


THANK YOU